

Chemistry, 2018 EC, entrance exam, version 5

Grade 12 · Chemistry · 2018 E.C.

Time allowed: 120 minutes**Questions:** 77**Total marks:** 77

Name _____ Grade _____ ID No. _____ Date _____

1. During discovered nucleus by Rutherford using alpha particles and gold foil experiment, why Majority of particles pass through the gold foil? Because (1)
- A. The atom has a positive charge
 - B. In the nucleus, is a dense central core within the atom.
 - C. In the nucleus there is heavy positively charged core
 - D. Most of the space in the atom is empty
-
2. A student is studying an atom with an atomic number of 31 and mass number of 70. Based on nuclear composition, which of the following statements is CORRECT? (1)
- A. The nucleus contains 31 protons and 39 neutrons
 - B. The nucleus contains 31 neutrons and 39 protons
 - C. The nucleus contains only 31 protons
 - D. The nucleus contains 70 neutrons and 31 protons
-
3. Naturally occurring boron consists of two isotopes, ^{10}B and ^{11}B , with the isotopic masses 10.010 amu and 11.01 amu, respectively. The observed average atomic mass of boron is 10.811 amu. What is the percentage abundance of ^{10}B and ^{11}B isotope respectively? (1)
- A. 11.1% and 88.90%
 - B. 19.90% and 80.10%
 - C. 10.10% and 89.90%
 - D. 22.2% and 77.80%
-
4. Which of the following atomic models is expressed CORRECTLY? (1)
- A. Dalton's atomic model — billiard ball model
 - B. Bohr model of atom — Plum pudding model
 - C. Thomson's model of atom — Planetary model
 - D. Rutherford atomic model — Ice cream model
-
5. Suppose an unknown element does NOT deflect cathode rays in an electric field but has significant mass. Based on atomic models, which subatomic particle could it be composed of, and how would this impact the element's atomic structure? (1)
- A. Protons; making the nucleus positively charged
 - B. Electrons; moving freely in orbitals
 - C. Neutrons; contributing to atomic mass without charge
 - D. Positrons; leading to an unstable atom
-
6. When elements are arranged in the order of increasing atomic masses every eight elements exhibits similar properties, much like the musical octave. This statement was stated by? (1)
- A. Dobernier
 - B. Mosley
 - C. Mendeleev
 - D. Newland

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7. Which of the following is NOT the key role of periodicity in the context of the periodic table? (1)
- A. to predicting element properties B. to classifying elements
C. to predicting electromagnetic radiation D. to provides the reactivity of elements
-
8. An element having the valence electron configuration $1s^2 2s^2 2p^6 3s^2 3p^4$ belongs to group? (1)
- A. IIA B. VIA
C. IVA D. VIIIA
-
9. Which of the following is the correct order of the atomic size of Mg^{2+} , C^{4-} , Na^+ , N^{3-} , and O^{2-} ? (1)
- A. $Na^+ < Mg^{2+} < C^{4-} < N^{3-} < O^{2-}$ B. $C^{4-} > N^{3-} > O^{2-} > Na^+ > Mg^{2+}$
C. $Mg^{2+} > Na^+ > O^{2-} > N^{3-} > C^{4-}$ D. $N^{3-} > O^{2-} > C^{4-} > Na^+ > Mg^{2+}$
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10. What types of bond is formed by sharing of electrons between atoms? (1)
- A. ionic bond B. metallic bond
C. electrovalent bond D. covalent bond
-
11. Which one of the following is responsible for the unusual high boiling points of HF, H₂O and NH₃? (1)
- A. hydrogen bond B. covalent bond
C. London dispersion force D. ionic bond
-
12. Which of the following compounds would most likely conduct electricity when dissolved in water but not in solid? (1)
- A. SiO₂ B. Cu
C. NaCl D. C₁₂H₂₂O₁₁
-
13. Which of the following is CORRECT about the formation of an ionic bond? An ionic bond is formed by: (1)
- A. mobile and delocalized electrons
B. By sharing of electrons
C. between positively and negatively charged ions
D. between stationary electrons
-
14. What property makes metallic elements are malleable means hammered into sheet? (1)
- A. its good conductivity B. its delocalized electrons
C. its strength D. its hardness
-
15. Which one of the following compound contains both ionic and covalent bonds? (1)
- A. ICl₃ B. CaO
C. Al₂O₃ D. Na₂CO₃
-
16. Which of the following intermolecular forces is generally the strongest? (1)
- A. London dispersion forces B. hydrogen bonding
C. dipole-dipole forces D. ion-dipole interaction
-
17. Consider the following chemical reaction: $2Mg(NO_3)_2 \rightarrow 2MgO + 4NO_2 + O_2$. The chemical reaction that takes place during the conversion of $Mg(NO_3)_2$ into the product is classified as _____ (1)
- A. Single replacement reaction B. Double displacement reaction
C. Decomposition reaction D. Combination reaction

-
18. What is the primary purpose of balancing chemical equation? (1)
- A. to make equation neater and more symmetrical
 - B. to satisfy the law of conservation of mass
 - C. to indicate physical state
 - D. to show the rate of reaction
-
19. Which one of the following is CORRECT about redox reaction? (1)
- A. Electrons are transferred between species bonding partners
 - B. Acid and base neutralize each other
 - C. atoms or ions of two compounds exchange
 - D. two soluble ions in compounds form insoluble products
-
20. A compounds found to contain 40g carbon, 6.7g hydrogen, 53.3g oxygen by mass. What is empirical formula of a compound? (1)
- | | |
|----------------------|---|
| A. CH ₄ O | B. C ₂ H ₄ O ₂ |
| C. CHO | D. CH ₂ O |
-
21. When 4g of magnesium nitride is react with excess water 3.2g of magnesium oxide is produced. $\text{Mg}_3\text{N}_2 + 3\text{H}_2\text{O} \rightarrow 2\text{NH}_3 + 3\text{MgO}$. What is percent yield of this reaction? (1)
- | | |
|-----------|----------|
| A. 66.67% | B. 94.5% |
| C. 78.6% | D. 49.4% |
-
22. Given the following equation representing the reaction: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$. How many grams of SO₃ will be produced when 6.8L oxygen is react with sulfur dioxide? (1)
- | | |
|---------|-----------|
| A. 68g | B. 48.57g |
| C. 160g | D. 4.76g |
-
23. What volume of oxygen at STP is required to burn 2 moles of propane (C₃H₈)? $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{g})$ (1)
- | | |
|----------|----------|
| A. 44.8L | B. 112L |
| C. 224L | D. 22.4L |
-
24. Which of the following is NOT a characteristic of heterogeneous mixtures? (1)
- A. They are uniform in appearance and properties
 - B. They can have varying compositions in different parts
 - C. Components are easily distinguishable
 - D. They can contain solid, liquid, and gas phases simultaneously
-
25. Which of the following statement is best describes about the heat of solution? (1)
- A. The heat required to change a solid to a liquid
 - B. The heat generated during a chemical reaction
 - C. The heat required to vaporize a liquid
 - D. The heat absorbed or released when a solute dissolves in a solvent
-
26. A laboratory technician prepares a solution by mixing 20 g of CaCl₂ (molar mass = 110.98 g/mol) in 300 g of water. What is the molality of the resulting solution? (1)
- | | |
|-----------|-----------|
| A. 0.75m | B. 0.50 m |
| C. 0.60 m | D. 0.70 m |

37. Which one of the following electromagnetic radiations has the shortest wavelength? (1)
- A. X-rays
B. gamma rays
C. UV rays
D. microwaves
-
38. What is energy of hydrogen atom at $n=1$? (1)
- A. -2.18×10^{-18} J
B. 4.38×10^{-18} J
C. $+2.18 \times 10^{-18}$ J
D. 1.5×10^{-19} J
-
39. Which of the following quantum numbers are described about the electron size, subshell and orientation of electrons atom respectively? (1)
- A. n, l, m_s
B. n, m_l, l
C. n, m_s, l
D. n, l, m_l
-
40. Which one of the following is not permissible about quantum numbers? (1)
- A. $n=3, l=2, m_l=0$
B. $n=5, l=3, m_l=1$
C. $n=1, l=0, m_l=0$
D. $n=4, l=5, m_l=-3$
-
41. The principle state, 'The sequence order in which electrons are added to atoms orbitals starting from the lowest energy level and progressing to higher one' is proposed by: (1)
- A. Hund's rule
B. Pauli's exclusion principle
C. Aufbau principle
D. Bohr principle
-
42. Which of the following is the correct electron configuration of Copper element at ground state? (Atomic number of Cu = 29) (1)
- A. $[\text{Ar}]4s^23d^{10}$
B. $[\text{Ar}]4s^13d^{10}$
C. $[\text{Ar}]4s^13d^5$
D. $[\text{Ar}]4s^23d^9$
-
43. Which one of the following compound contains both ionic and covalent bonds? (1)
- A. Na_2CO_3
B. CaO
C. Al_2O_3
D. ICl_3
-
44. Among the following which one is NOT TRUE about ionic compounds? (1)
- A. ionic compounds are aggregate of ions
B. ionic compounds are usually soluble in water
C. ionic compounds have high melting point
D. ionic compounds conduct electricity in any state
-
45. Water molecules have two bond pairs and two lone pairs of electrons. Which of the following is correct about these pairs of electrons? (1)
- A. The repulsion between two lone pairs is greater than the repulsion between the two bonding pairs
B. Lone pair's repulsion is exactly identical with bonding pair-bonding pair repulsion
C. The repulsion between two bonding pairs is greater than the repulsion between the two lone pairs
D. Bonding pair's repulsion is exactly identical with lone pair-bonding pair repulsion
-
46. What is the hybridization of carbon atom in C_2H_2 ? (1)
- A. sp^3d
B. sp
C. sp^2
D. sp^3

47. Which of the following molecule does NOT obey the octet rule? (1)
- A. CS₂ B. PCl₃
C. AlF₃ D. HCN
48. Which of the following molecules has a trigonal bipyramidal structure based on electron set arrangement? (1)
- A. IF₅ B. ICl₄⁻
C. BrF₅ D. SF₄
49. Which of the following properties is associated with covalent compounds? Covalent Compounds are: (1)
- A. good electrical conductivity B. form ions in the solution
C. crystalline solid at room temperature D. exist as solid, liquid or gas state
50. Which of the following is NOT TRUE about the physical properties of metals? (1)
- A. Metals have high electrical and thermal conductivity
B. Metals are malleable and ductile
C. Metals are not durable
D. Metals are very cool and shiny
51. Which hybridization changes when the carbon atom undergoes in the combustion of methane? CH₄(g) + 2O₂(g) → CO₂(g) + 2H₂O(g) (1)
- A. sp³ → sp³ B. sp³ → sp²
C. sp² → sp D. sp³ → sp
52. Which of the following represents the correct molecular orbital electron configuration of O₂? (1)
- A. σ1s² σ*1s² σ2s² σ*2s² σ2p_x² (π2p_y² = π2p_z²) (π*2p_y¹ = π*2p_z¹)
B. σ1s² σ*1s² σ2s² σ*2s² σ2p_x² π2p_y² = π2p_z² π*2p_y²
C. σ1s² σ*1s² σ2s² σ*2s² π2p_y² = π2p_z² σ2p_x² π*2p_y² = π*2p_z¹
D. σ1s² σ*1s² σ2s² σ*2s² σ2p_x² π2p_y² = π2p_z² π*2p_y²
53. Which of the following statement is TRUE about paramagnetic substances? (1)
- A. they have all pair's electrons
B. they are attracted to magnetic fields due to unpaired electrons
C. they are always solid at room temperature
D. they have low boiling point than diamagnetic substances
54. What is the bond order of diatomic molecule N₂? (1)
- A. 2.5 B. 1.5
C. 3 D. 2
55. Which of the following substances is example of molecular crystal? (1)
- A. NaCl B. SiO₂
C. Fe D. C₆H₁₂O₆
56. From the given assumption of kinetic molecular theory of gases which of the following is CORRECT? (1)
- A. A gas is a collection of particles in constant, curved-line motion.
B. There is no space between gas particles compared with the size of the particles themselves
C. Gas particles do not attract or repel each other, they do not interact
D. The average kinetic energy of gas particles is inversely proportional to the temperature of the gas in kelvins.

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57. The molar volume of a gas is: (1)
- A. The 1 mole of a gas occupied by 22.4 liters at standard temperature and pressure
 - B. The volume occupied by 2 mole of a gas at a given temperature
 - C. The volume of a gas having weight of 1g
 - D. The volume equivalent to 1000ml of a gas at 273K and 1atm pressure
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58. Which of the following is CORRECTLY describes conjugate acids and conjugate bases? (1)
- A. Conjugate base has one more hydrogen and one fewer minus charge than the corresponding base
 - B. Conjugate base is a species formed after removal of proton from an acid whereas conjugate acid is formed after addition of proton to a base
 - C. Conjugate acid has one more fewer and one more minus charge than the corresponding base
 - D. Conjugate acid is a species formed after removal of proton from a base whereas conjugate base is formed after addition of proton to an acid
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59. Which of the following molecule is act as a Lewis acid? (1)
- | | |
|------------------|------------------|
| A. BF_3 | B. NH_3 |
| C. Cl^- | D. OH^- |
-
60. The pH of a 0.1M solution of aqueous solution of certain acid is 5. What is the value of acid ionization constant (K_a) of this acid? (1)
- | | |
|--------------------------|-------------------------|
| A. 1.0×10^{-10} | B. 1.0×10^{-5} |
| C. 1.0×10^{-4} | D. 1.0×10^{-9} |
-
61. What happen to the degree of ionization of 0.1M solution of acetic acid (CH_3COOH) if small amount of sodium acetate (CH_3COONa) is added? (1)
- A. the degree of ionization increase because the concentration of ions increases
 - B. the degree of ionization decreases because the equilibrium shift toward the reactants
 - C. the degree of ionization remains constant because K_a is constant
 - D. the pH value of the solution depends on position of equilibrium
-
62. What is the pH of a buffer solution containing 0.15 M acetic acid (CH_3COOH) and 0.10 M sodium acetate (CH_3COONa)? (Given $\text{p}K_a$ of acetic acid = 4.74) (1)
- | | |
|---------|----------|
| A. 4.56 | B. 10.76 |
| C. 5.06 | D. 5.26 |
-
63. Which of the following salts would likely produce an acidic solution when dissolved in water? (1)
- | | |
|------------------------------|----------------------------|
| A. CH_3COONa | B. K_2CO_3 |
| C. NH_4Cl | D. CaSO_4 |
-
64. What is the primary difference between the endpoint and the equivalence point in a titration? (1)
- A. The endpoint is where the reaction is complete, while the equivalence point is where the indicator changes color.
 - B. The endpoint is where the indicator changes color, while the equivalence point is where the amount of titrant equals the amount of analyte
 - C. The endpoint occurs before the equivalence point.
 - D. There is no difference; they are synonymous terms.

65. What is oxidation number of Fe in the $K_4[Fe(CN)_6]$? (1)

- A. +2
B. +6
C. -4
D. +1

66. In the reaction $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$, which species is oxidized? (1)

- A. NaOH
B. H₂O
C. Na
D. H₂

67. Which one of the following is TRUE in the electrolysis of dilute H_2SO_4 solution? (1)

- A. oxygen gas is liberated at cathode electrode
B. concentration of H_2SO_4 is increase after electrolysis
C. hydrogen gas is released at anode electrode
D. water is not undergo electrolysis

68. One of the basic differences between metallic conduction and electrolytic conduction is: (1)

- A.** The charge carriers in metals are localized electrons while strongly held ions are in electrolytic conductors
- B.** The conduction of electricity is due to the movement of ions in electrolytic conduction while the flow of electrons in metallic conduction
- C.** The charge carriers in metals are cations and anions while mobile electrons are in electrolytic conduction
- D.** The conduction of electricity is due to the movement of ions in metallic conduction while the flow of electrons in electrolytic conduction

69. Three electrolytic cells are connected in series that contain Au^{3+} , Ag^{+} and Cu^{2+} ions. How many grams of Ag^{+} and Cu^{2+} metal will be deposited at cathode if 28g of Au^{3+} is deposited respectively? (1)

- A. 65.66 g and 32g
B. 25.33 g and 7.51g
C. 108g and 14g
D. 46.05g and 13.64g

70. In the electrolysis of a concentrated solution of sodium chloride (brine), which gas is produced at the anode? (1)

- A.** Chlorine **B.** Hydrogen
C. Oxygen **D.** Nitrogen

71. In a galvanic cell, how can you identify the cathode when given standard reduction potentials? (1)

- A. The electrode that is larger in size is the cathode
- B. The electrode with the lower reduction potential is the used as electrolyte
- C. The electrode with the higher reduction potential is the cathode
- D. The electrode that is connected to the positive terminal of the battery is always the cathode

72. Which of the following statements is TRUE regarding the half-reactions in a galvanic cell? (1)

- A.** The anode half-reaction involves loss of electrons
- B.** The cathode half-reaction involves oxidation
- C.** The cathode half-reaction produces ions
- D.** The anode half-reaction involves reduction

73. Given the reaction: $\text{Al(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Al}^{3+}(\text{aq}) + \text{Cu(s)}$: which of the following is the correct cell notation? (1)

- A. $\text{Cu}^{2+}(\text{aq}) / \text{Cu}(\text{s}) // \text{Al}^{3+}(\text{aq}) / \text{Al}(\text{s})$
 B. $\text{Al}(\text{s}) / \text{Al}^{3+}(\text{aq}) // \text{Cu}(\text{s}) / \text{Cu}^{2+}(\text{aq})$
 C. $\text{Cu}^{2+}(\text{aq}) / \text{Cu}(\text{s}) // \text{Al}(\text{s}) / \text{Al}^{3+}(\text{aq})$
 D. $\text{Al}(\text{s}) / \text{Al}^{3+}(\text{aq}) // \text{Cu}^{2+}(\text{aq}) / \text{Cu}(\text{s})$

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- 74.** Which one of the following are renewable natural resources? (1)
- | | |
|---------------------|---------------------|
| A. Animals | B. Coal |
| C. petroleum | D. crude oil |
-
- 75.** The process of producing ammonia in the industry is by: (1)
- | | |
|---------------------------|---------------------------|
| A. Contact process | B. Ostwald process |
| C. Solvay process | D. Haber process |
-
- 76.** Which one of the following is INCORRECT about chemical industries? Chemical industry is (1)
- A.** use safe operation methods in their manufacturing process
 - B.** consume relatively large quantities of energy during production
 - C.** use only naturally available raw materials
 - D.** test their products before and after production
-
- 77.** What is one of the primary industrial uses of ammonia? (1)
- | | |
|---|--|
| A. As a food preservative | B. In the production of fertilizers |
| C. As a cleaning agent for glass | D. In the manufacturing of plastics |

— END OF EXAMINATION —